

Review retrospective:
*the generic- (k_1, k_2) moment lift — the 2D trio reviewed
end-to-end*

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Abstract

A soundness review of `Laplace/TwoD/KthKthMomentAsymptotic.lean`: the generic- (k_1, k_2) asymptotic of the 2D separable mixed-even Gibbs moment $\langle x^{2j_1} y^{2j_2} \rangle_{2D}$, the third and final leg of the “2D lift” trio. Result: **a clean fidelity pass with one dead-code removal** (the dead `open Real`). With this, the trio — partition, numerator, moment — and their factorisation root are all reviewed.

1 Setting

This is the Gibbs-moment leg of the lift pattern: it factorises the 2D mixed moment as a product of two 1D even moments (via `AddSeparable`’s `gibbsExpectation_separable_addSeparable`), substitutes the 1D closed form `gibbsExpectation_kthPotential_even`, and packages the limit. It is the moment quotient of the two legs reviewed just before it (partition, numerator).

2 What was reviewed

Four public theorems and the composite constant $C(k_1, k_2, j_1, j_2) = (2k_1)!^{j_1/k_1} (2k_2)!^{j_2/k_2} \frac{\Gamma((2j_1+1)/(2k_1))}{\Gamma(1/(2k_1))} \frac{\Gamma((2j_2+1)/(2k_2))}{\Gamma(1/(2k_2))}$: the factorisation $\langle \cdot \rangle_{2D} = \langle \cdot \rangle_U \langle \cdot \rangle_V$, the closed form $C \cdot t^{-j_1/k_1 - j_2/k_2}$, the rescaled limit $\rightarrow C$, and the asymptotic equivalence.

3 Fidelity / soundness findings

Sound; no mismatch. The constant is the product of the two 1D `gibbsExpectation_kthPotential_even` constants, and the per-coordinate exponent $-j/k$ is exactly the moment relation: numerator exponent $-(2j+1)/(2k)$ minus partition exponent $-1/(2k)$. The factorisation discharges the four integrability obligations and both $Z \neq 0$ conditions from the 1D infrastructure. Hypotheses sit correctly ($t > 0$ exact, $k_i \geq 1$ for the limit). Docstrings match the statements — like the numerator leg, and unlike the partition leg whose summary bullet had mislabelled a theorem.

4 Simplifications applied

One dead open. `open Real` is dead (all `Real.*` qualified, bare t^{real} is `rpow`); `MeasureTheory` (bare `Integrable`), `Filter`, `Asymptotics`, scoped `Topology` stay. Module green at 2700 jobs, no warnings, signatures byte-identical. GPT affirmed.

5 Disagreements and open questions

None. The reviewer's other suggestions (collapse the $n = 0$ integrability sub-proofs to `simpa`; replace the inline `show ... by ring` equality proofs inside `rw`) were declined as proof-style refactors, consistent with the two sibling legs.

6 Lessons

- **An end-to-end family review converges fast.** Reviewing the four files — factorisation root, then partition, numerator, moment legs — in dependency order made each fidelity check incremental: only the per-file constant and exponent were new, and the dead `open Real` was found in three of the three legs (the root used `open Real MeasureTheory` too, also trimmed). One genuine docstring drift surfaced across the four (the partition bullet), and the rebuild settled the one wrong simplification (the root's transitively-load-bearing `SemiDegenerate import`).
- **The moment exponent is a checkable arithmetic relation.** $-j/k = (-(2j+1)/(2k)) - (-1/(2k))$ ties the moment leg's exponent to the numerator and partition legs already reviewed — a free cross-check that the three constants are mutually consistent.

7 Follow-ups

- `KthKthOddMoments` (the odd-moment vanishing) and the $(2,3)$ `QuarticSextic*` specialisations remain — the latter near-certain repeats of the dead `open Real`.
- Proof-golf forward tide (shared across the trio): the `simpa` integrability shortenings and the `h_k1/h_k2`-style dedup helper.